10–12 Minute Presentation Guide & Speaker Notes – FinDoc-AuditEngine v3.0

Companion Files in docs/:

- PowerPoint Presentation (.pptx): [FinDoc_AuditEngine_Presentation_Slides.pptx](./FinDoc_AuditEngine_Presentation_Slides.pptx) (12 Widescreen 16:9 Slides with Speaker Notes)
- Landscape Slide Deck (.pdf): [FinDoc_AuditEngine_Slide_Deck.pdf](./FinDoc_AuditEngine_Slide_Deck.pdf) (12 Designed 16:9 Slides)
- Recorded Demo Walkthrough Video (.mp4): [FinDoc_AuditEngine_Demo_Walkthrough.mp4](./FinDoc_AuditEngine_Demo_Walkthrough.mp4) (HD 720p Walkthrough of Dashboard, 3 Scenarios, Terminal Logs & Tests)

Recommended 10–12 Minute Presentation Timeline

Time Segment	Duration	Presentation Section	Slides Covered	Key Demonstration / Narrative Focus
Part 1	1 Min (00:00–01:00)	Problem & Solution	Slides 1–2	Why 3-Way Invoice–PO–GRN reconciliation, regex PII masking, and anomaly scoring are useful together.
Part 2	2 Min (01:00–03:00)	System Architecture	Slides 3–5	FastAPI pipeline: regex privacy masking, deterministic 3-way matcher, seeded synthetic Isolation Forest (N=1,000), approval router, and in-memory prototype ledger.
Part 3	5 Min (03:00–08:00)	Live Demonstration (3 Scenarios)	Slides 6–8	Live walkthrough of Scenario 1 (INV-1001 Clean Match), Scenario 2 (INV-1004 Price Discrepancy), and Scenario 3 (INV-1005 Critical-Risk Anomaly) + terminal output.
Part 4	2 Min (08:00–10:00)	Testing & Challenges	Slides 9–10	Show python -m pytest tests -q (28 passed in Day 29 / 33 passed in Day 30 – all collected tests passed) and explain the DOM decoupling & scaling fixes.
Part 5	1–2 Min (10:00–12:00)	Limitations & Future Work	Slides 11–12	Discuss in-memory ledger limitation, synthetic ML baseline calibration needs, security/RBAC controls, and PostgreSQL/OCR roadmap.

Detailed Speaker Script by Section

1. Problem and Solution (1 Minute – Slides 1–2)

- *"Hello everyone. Today I am presenting FinDoc-AuditEngine v3.0, our internship capstone prototype for Accounts Payable reconciliation and anomaly detection."*
- *"In procurement, finance teams must verify that every Vendor Invoice matches both the contractual Purchase Order (PO) and the warehouse Goods Receipt Note (GRN). Pure rule-based checks catch explicit unit-price or quantity mismatches, but they cannot flag an invoice that matches a PO on paper while exhibiting unusual risk signals–such as an offshore vendor demanding a 1-day rush payment just below a $50,000 sign-off limit."*
- *"Our prototype addresses this by combining pre-inference regex PII masking, deterministic 3-way SKU reconciliation, and a 7-feature Isolation Forest anomaly scorer with local perturbation explanations."*

2. Architecture (2 Minutes – Slides 3–5)

- *"Every request to POST /api/v1/audit/process passes through four modular stages:"*

1. *"First, FinancialPrivacyScrubber applies 6 regex patterns to mask sensitive identifiers like Indian PAN cards, Aadhaar-formatted numbers, bank accounts, phones, and emails before logging."*
2. *"Second, ReconciliationMatcher performs deterministic 3-way matching across SKU unit prices and net accepted dock quantities (received minus rejected) within a $0.05 subtotal tolerance."*
3. *"Third, AnomalyScorer transforms 7 engineered features through StandardScaler and a 200-tree IsolationForest trained on a reproducible seeded synthetic baseline of 1,000 records. It outputs a 0–100 engineered anomaly-risk indicator and calculates single-feature perturbation attributions."*
4. *"Fourth, ApprovalRouter assigns the invoice to one of five governance queues and records the decision in our prototype's in-memory AUDIT_LEDGER list, displaying totals in both USD and INR (1 USD = 86.50 INR)."*

3. Live Demonstration – Three Representative Scenarios (5 Minutes – Slides 6–8)

- Scenario 1 – Clean Invoice (INV-1001, OfficeDepot India Pvt Ltd, $432.00 / INR 37,368.00):
- *"Let's load Scenario 1 in the dashboard. Two ergonomic chairs are billed at $180.00 each, matching PO-9001 exactly with zero dock rejections. Price variance is $0.00, the engineered risk indicator is low (25.0/100), and the router assigns TIER_1_STP_AUTO_APPROVED."*
- Scenario 2 – Price Discrepancy (INV-1004, Overcharge Technologies, $5,400.00 / INR 467,100.00):
- *"Next, let's test Scenario 2. Here, 10 enterprise SSD arrays are billed at $450.00/unit, whereas PO-9004 specifies $400.00/unit. The deterministic 3-way matcher immediately flags RATE_VARIANCE with a +$500.00 (INR 43,250.00) overcharge and blocks auto-approval in REJECTED_DISCREPANCY. We can also edit any cell in the line-item table live to see variances recompute dynamically."*
- Scenario 3 – Critical-Risk Anomaly (INV-1005, Shadow Island Holding BZ, $49,500.00 / INR 4,281,750.00):
- *"Finally, Scenario 3 demonstrates why we pair rules with anomaly scoring. INV-1005 bills $49,500 and matches PO-9005 on price and quantity. However, the 7-D Isolation Forest scores it at 99.0/100 (CRITICAL) because of the combination of an offshore jurisdiction (BZ), a Net-1 rush payment window, and an amount just under $50,000. The perturbation bars show those exact features driving the score, and the router places the invoice in FROZEN_FRAUD_RISK."*

4. Testing and Engineering Challenges (2 Minutes – Slides 9–10)

- *"In the terminal, running python -m pytest tests -q verifies that all collected tests pass (28 passed in Day 29 and 33 passed in Day 30), covering matching math, regex masking, error handling, and dynamic multi-line API requests."*
- *"Two key engineering challenges we solved were: (1) decoupling the frontend line-item table from static preset objects via collectLineItemsFromDOM() so every user edit is evaluated dynamically, and (2) applying $\ln(1 + \text{Total}_{\text{USD}})$ and StandardScaler so large dollar amounts do not overshadow bounded variance ratios."*

5. Limitations and Future Work (1–2 Minutes – Slides 11–12)

- *"To conclude, we want to be clear about the current prototype's limitations and how we would evolve it for production:"*

1. *"First, AUDIT_LEDGER is currently an in-memory Python list that resets on restart; production requires durable PostgreSQL storage and Redis idempotency keys."*
2. *"Second, our IsolationForest is fit on a seeded synthetic baseline of 1,000 records, and its 0–100 score is an uncalibrated anomaly indicator. Production deployment requires evaluation and threshold calibration on authorized historical company invoices plus drift monitoring."*
3. *"Third, /api/v1/audit/parse-raw performs regex invoice-text parsing; integrating optical OCR (LayoutLMv3) and OAuth2/RBAC security controls are key next steps."*